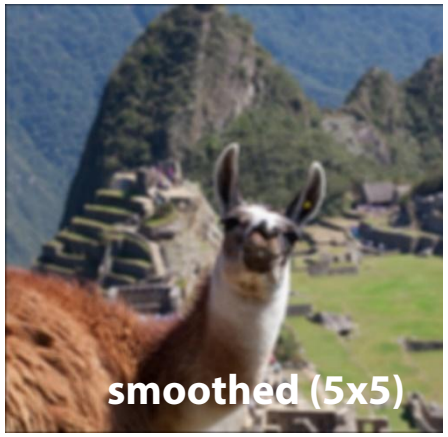
Sharpening revisited

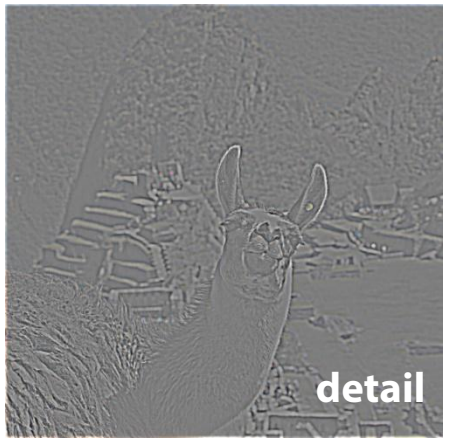
- What does blurring take away?



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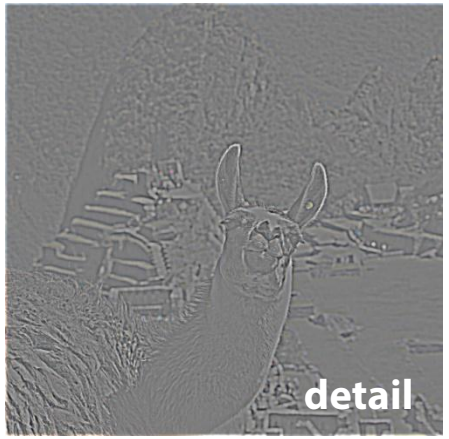


(This "detail extraction" operation is also called a ***high-pass filter***)

Let's add it back:



+ α



=



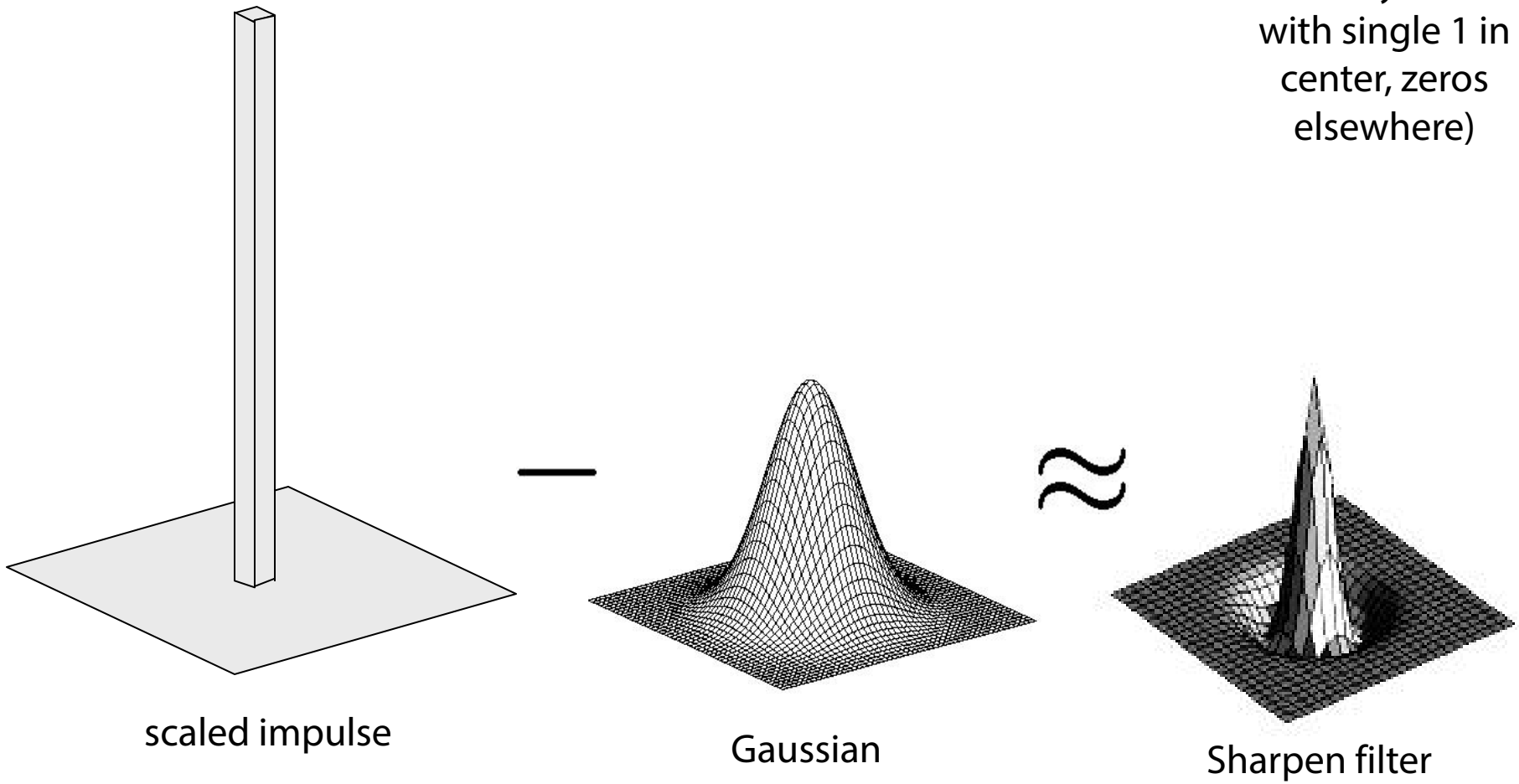
Photo credit: https://www.flickr.com/photos/geezaweezer/16089096376/

Sharpen filter

$$F + \alpha\,(F - \underbrace{F * H}_{\text{blurred image}}) =$$

↑ image

↑ unit impulse (identity kernel with single 1 in center, zeros elsewhere)

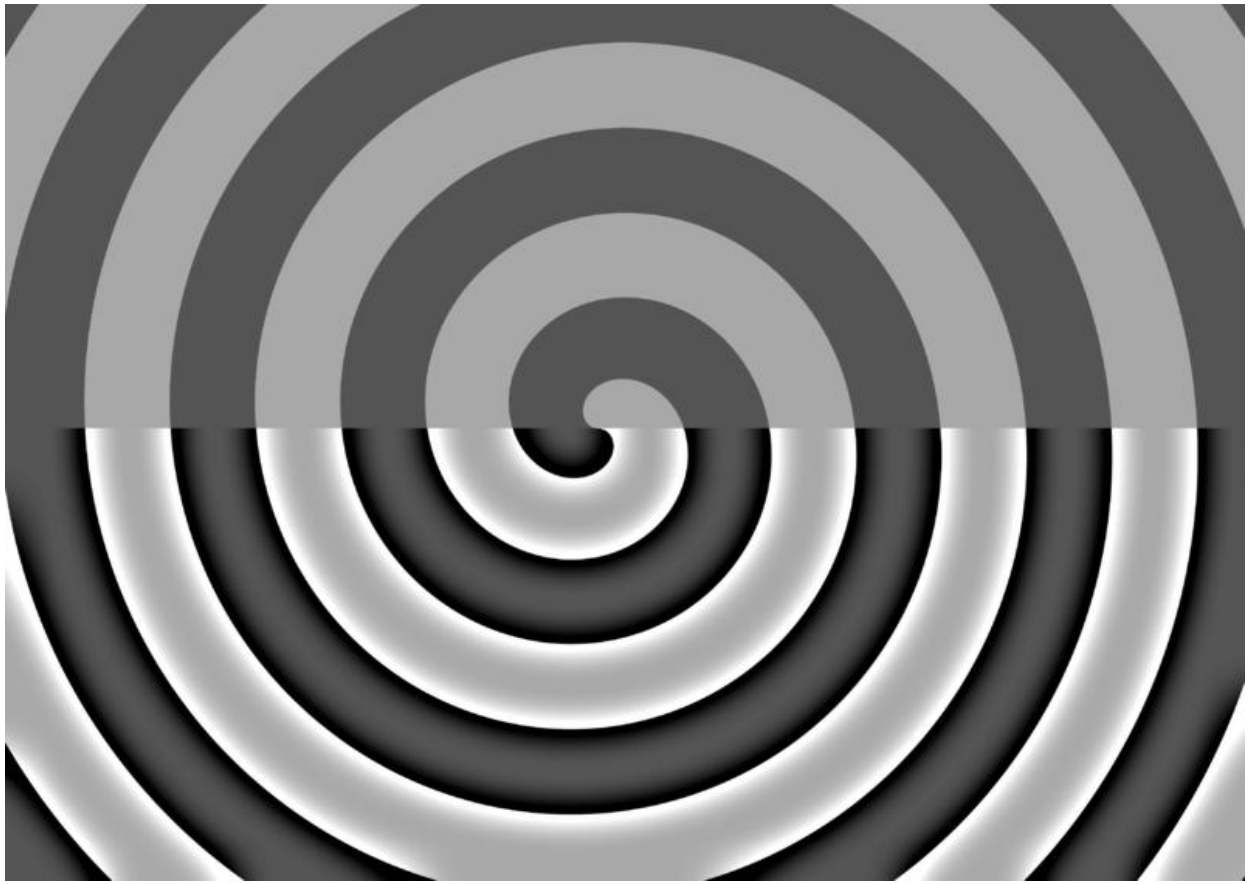


scaled impulse — Gaussian ≈ Sharpen filter

Sharpen filter

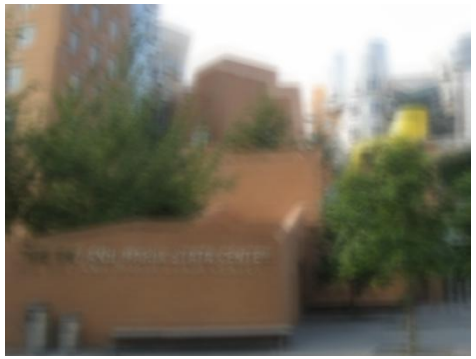
unfiltered

filtered



“Optical” convolution

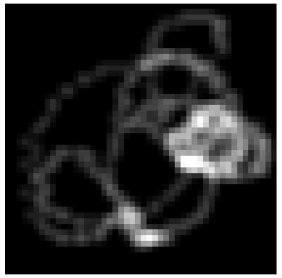
Camera shake



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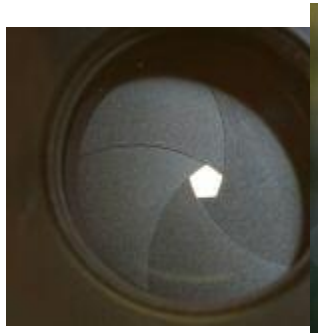


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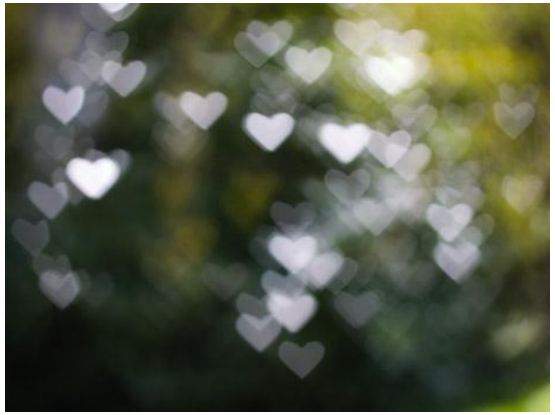
Source: Fergus, *et al.* “Removing Camera Shake from a Single Photograph”, SIGGRAPH 2006

Bokeh: Blur in out-of-focus regions of an image.









Source: https://www.diyphotography.net/diy_create_your_own_bokeh/

Filters: Thresholding





$$g(m,n) = \begin{cases} 255, & f(m,n) > A \\ 0 & otherwise \end{cases}$$

Linear filters

- Can thresholding be implemented with a linear filter?

Questions?

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